

# PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

## Module 6 Exam Notes

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## 1. Uniform Linear Arrays (ULA)

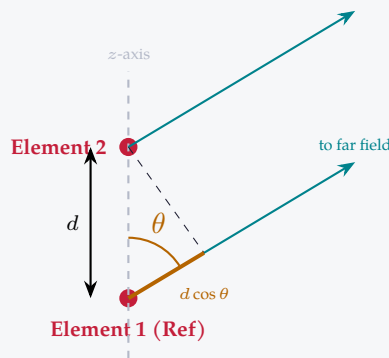
### Definition of an Antenna Array

An **antenna array** is a set of two or more identical antenna elements positioned and excited such that their individual radiation fields combine constructively in desired directions and cancel destructively in other directions, increasing directivity.

### 1.1 Two-Element Array Analysis

Consider two identical antennas separated by a distance  $d$  along the  $z$ -axis. Antenna 1 is at the origin and serves as the phase reference ( $I_1 = I_0 e^{j0}$ ). Antenna 2 is fed with a phase lead  $\beta$  ( $I_2 = I_0 e^{j\beta}$ ).

#### Two-Element Array Geometry



In the far field, the path difference between the waves is  $d \cos \theta$  (where  $\theta$  is measured from the array axis). The total phase difference  $\psi$  between the fields is:

$$\psi = kd \cos \theta + \beta$$

The total far-field electric field is:

$$E_{total} = E_1 + E_2 = E_0 (1 + e^{j(kd \cos \theta + \beta)}) = E_0 e^{j\psi/2} \left( 2 \cos \frac{\psi}{2} \right)$$

The normalized **Array Factor (AF)** is:

$$AF_n(\theta) = \cos \left( \frac{kd \cos \theta + \beta}{2} \right)$$

### 1.2 N-Element Uniform Linear Array (ULA)

For  $N$  identical elements fed with equal amplitude  $I_0$  and progressive phase shift  $\beta$ :

$$AF = \sum_{n=0}^{N-1} e^{jn(kd \cos \theta + \beta)} = \sum_{n=0}^{N-1} e^{jn\psi}$$

Summing this geometric progression:

$$|AF| = \left| \frac{\sin \left( \frac{N}{2} \psi \right)}{\sin \left( \frac{1}{2} \psi \right)} \right| \Rightarrow |AF_n| = \left| \frac{\sin \left( \frac{N}{2} (kd \cos \theta + \beta) \right)}{N \sin \left( \frac{1}{2} (kd \cos \theta + \beta) \right)} \right|$$

### 1.3 Broadside vs. End-Fire Arrays

**1. Broadside Array:** Maximum radiation is perpendicular to the array axis ( $\theta = 90^\circ$ ).

- *Condition:*  $\psi = 0$  at  $\theta = 90^\circ \Rightarrow kd \cos(90^\circ) + \beta = 0 \Rightarrow \beta = 0$ .
- *Nulls:* Occur when  $\sin(N\psi/2) = 0 \Rightarrow \theta_{null} = \cos^{-1}(\pm \frac{m\lambda}{Nd})$ ,  $m = 1, 2, 3, \dots$

**2. End-Fire Array:** Maximum radiation is along the array axis ( $\theta = 0^\circ$  or  $180^\circ$ ).

- *Condition for  $\theta = 0^\circ$ :*  $\psi = 0 \Rightarrow kd \cos(0^\circ) + \beta = 0 \Rightarrow \beta = -kd$ .
- *Condition for  $\theta = 180^\circ$ :*  $\psi = 0 \Rightarrow kd \cos(180^\circ) + \beta = 0 \Rightarrow \beta = kd$ .

## 2. Non-Uniform Amplitude Arrays

Uniform arrays have high directivity but suffer from relatively high sidelobes ( $-13.5$  dB for the first sidelobe). To reduce sidelobes, elements can be excited with non-uniform current amplitudes.

### 2.1 Binomial Array

The excitation amplitudes of the elements correspond to the coefficients of a binomial expansion:

$$a_n = \binom{N-1}{n}, \quad n = 0, 1, \dots, N-1$$

For example, for a 5-element array, the amplitudes scale as  $1 : 4 : 6 : 4 : 1$ .

- **Advantage:** Completely eliminates sidelobes if the element spacing  $d \leq \lambda/2$ .
- **Disadvantage:** Broader beamwidth (HPBW) and lower directivity than a uniform array of equivalent length.

### 2.2 Dolph-Tschebyscheff Array

This array uses Tschebyscheff (Chebyshev) polynomials  $T_m(z)$  to design a pattern with a specified sidelobe level (SLL) and the narrowest possible main beamwidth. The Chebyshev polynomials are defined by:

$$T_m(z) = \cos(m \cos^{-1} z) \quad (|z| \leq 1)$$

$$T_m(z) = \cosh(m \cosh^{-1} z) \quad (|z| > 1)$$

This method optimizes the trade-off by ensuring that all sidelobes are at the exact same specified level (e.g.,  $-20$  dB or  $-30$  dB).

#### Design Procedure:

1. Specify the number of elements  $N$  (array order  $m = N - 1$ ) and the desired sidelobe level ratio  $R$  (e.g.,  $R = 26.3$  for  $-20$  dB SLL;  $R = 10^{SLL_{dB}/20}$ ).
2. Compute the parameter  $z_0$  that maps the main lobe maximum to  $T_m(z_0)$ :

$$z_0 = \cosh \left[ \frac{1}{m} \cosh^{-1}(R) \right]$$

3. Substitute  $z = z_0 \cos\left(\frac{\psi}{2}\right)$  into  $T_m(z)$  to obtain the array polynomial. The AF takes the form  $AF(\psi) = T_m\left(z_0 \cos\frac{\psi}{2}\right)$ .
4. Extract the excitation coefficients  $a_n$  by expanding the Chebyshev polynomial in powers of  $\cos^n\left(\frac{\psi}{2}\right)$ .

**Key result:** For a 5-element array with  $-20$  dB SLL, the excitation amplitudes are approximately  $1 : 2.41 : 3.14 : 2.41 : 1$  (compare to binomial  $1 : 4 : 6 : 4 : 1$  — Dolph gives a narrower beam for the same SLL).

### 3. Extension to Planar Arrays

A **planar array** consists of elements positioned in a two-dimensional grid (e.g.,  $M$  elements along  $x$  with spacing  $d_x$ , and  $N$  elements along  $y$  with spacing  $d_y$ ).

The overall Array Factor is the product of the individual linear array factors along the two directions:

$$AF_{planar} = AF_x \cdot AF_y = \left[ \frac{\sin\left(\frac{M}{2}\psi_x\right)}{\sin\left(\frac{1}{2}\psi_x\right)} \right] \left[ \frac{\sin\left(\frac{N}{2}\psi_y\right)}{\sin\left(\frac{1}{2}\psi_y\right)} \right]$$

where:

$$\psi_x = kd_x \sin \theta \cos \phi + \beta_x$$

$$\psi_y = kd_y \sin \theta \sin \phi + \beta_y$$

By dynamically adjusting the phase shifters  $\beta_x$  and  $\beta_y$ , the main beam can be steered to any coordinate  $(\theta_0, \phi_0)$  in 3D space:

$$\beta_x = -kd_x \sin \theta_0 \cos \phi_0$$

$$\beta_y = -kd_y \sin \theta_0 \sin \phi_0$$

### 4. Antenna Array Synthesis

Array synthesis involves finding the feeding currents (amplitudes and phases) and element locations required to produce a desired radiation pattern.

#### 4.1 Schelkunoff Polynomial Method

Schelkunoff represented the array factor of an  $N$ -element linear array as a polynomial in the complex variable  $z = e^{j\psi}$ :

$$AF = \sum_{n=0}^{N-1} a_n z^n = a_{N-1}(z - z_1)(z - z_2) \dots (z - z_{N-1})$$

where  $z_i$  are the roots (zeros) of the polynomial. Since  $z$  lies on the unit circle ( $|z| = 1$ ), the nulls in the radiation pattern correspond to the roots that lie on the unit circle.

**Visible Region:** As  $\theta$  varies from  $0^\circ$  to  $180^\circ$ ,  $\psi = kd \cos \theta + \beta$  traces an arc on the unit circle in the complex  $z$ -plane. This arc is called the **visible region**:

$$-kd + \beta \leq \psi \leq kd + \beta$$

- For  $d = \lambda/2$  and  $\beta = 0$ : visible region spans  $-\pi \leq \psi \leq \pi$  (full circle).
- For  $d < \lambda/2$ : visible region is less than a full circle — some roots in the **invisible region** do not contribute to the radiation pattern.
- Placing a root *inside* the visible region creates a null in the pattern. Placing a root *outside* the visible region eliminates that null.

This allows designers to synthesize specific patterns by precisely controlling which roots lie on vs. off the visible arc.

## 4.2 Woodward-Lawson Method

The Woodward-Lawson method synthesizes a desired source pattern by superimposing several independent, orthogonal beams (each having a  $\frac{\sin(N\psi/2)}{N \sin(\psi/2)}$  shape centred at different angles). The excitation coefficients are found by sampling the desired pattern at the beam centres.

**Sampling formula:** For  $N$  elements, the array factor is:

$$AF(\psi) = \sum_{m=-(N-1)/2}^{(N-1)/2} c_m \cdot \frac{\sin\left[\frac{N}{2}(\psi - \psi_m)\right]}{N \sin\left[\frac{1}{2}(\psi - \psi_m)\right]}$$

where each beam is centred at  $\psi_m = \frac{2\pi m}{N}$  and the sampling coefficients  $c_m$  are found by evaluating the desired pattern  $F_d(\psi)$  at those points:

$$c_m = F_d(\psi_m) = F_d\left(\frac{2\pi m}{N}\right)$$

### Key properties:

- The synthesized pattern passes exactly through the desired sample values.
- Between sample points, the approximation may deviate (Gibbs-like ripple for abrupt transitions).
- Adding more elements  $N$  increases the number of sample points and improves accuracy.
- Suitable for pencil-beam, sector, flat-top, or shaped beam synthesis.

## Quick Revision — Module 6

Antenna Arrays and Synthesis | PE-EC801A

### Key Formulas and Metrics at a Glance

|                                                      |                                                                                              |
|------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>N-Element ULA Array Factor (<math>AF</math>)</b>  | $AF = \frac{\sin(N\psi/2)}{\sin(\psi/2)}$ where $\psi = kd \cos \theta + \beta$              |
| <b>Broadside Phase Shift (<math>\beta</math>)</b>    | $\beta = 0$                                                                                  |
| <b>End-Fire Phase Shift (<math>\beta</math>)</b>     | $\beta = \mp kd$ (for $\theta = 0^\circ$ or $180^\circ$ main beam)                           |
| <b>Binomial Array Coefficients</b>                   | $a_n = \binom{N-1}{n} = \frac{(N-1)!}{n!(N-1-n)!}$                                           |
| <b>Planar Beam Steering Phase</b>                    | $\beta_x = -kd_x \sin \theta_0 \cos \phi_0, \quad \beta_y = -kd_y \sin \theta_0 \sin \phi_0$ |
| <b>Schelkunoff Complex Variable (<math>z</math>)</b> | $z = e^{j\psi} = e^{j(kd \cos \theta + \beta)}$                                              |

### One-Line Definitions

- **Array Factor:** The radiation pattern of an array of isotropic elements, describing how the geometry and feeds affect the pattern.
- **Grating Lobe:** An unwanted duplicate main beam of radiation that occurs when element spacing  $d \geq \lambda$ .
- **Binomial Array:** An array with non-uniform excitation coefficients that completely eliminates sidelobes.
- **Phased Array:** An array where the phase of each element is dynamically adjusted to steer the radiation beam without moving the antenna.
- **Woodward-Lawson Method:** A pattern synthesis technique based on the superposition of orthogonal beam patterns.

### Frequently Asked / Viva Questions

#### Q1. Define Array Factor ( $AF$ ). State the Principle of Pattern Multiplication.

The Array Factor is the radiation pattern of the array assuming isotropic radiating elements. The Principle of Pattern Multiplication states that the total far-field pattern ( $E_{total}$ ) of an array of identical non-isotropic elements is the product of the single-element field pattern ( $E_{single}$ ) and the Array Factor ( $AF$ ):

$$E_{total}(\theta, \phi) = E_{single}(\theta, \phi) \times AF(\theta, \phi)$$

#### Q2. Differentiate between broadside and end-fire arrays.

A broadside array radiates its maximum power perpendicular to the axis of the array ( $\theta = 90^\circ$ ), which requires in-phase feeds ( $\beta = 0$ ). An end-fire array radiates along the axis ( $\theta = 0^\circ$  or  $180^\circ$ ), which requires a progressive phase shift  $\beta = \mp kd$ .

## Frequently Asked / Viva Questions (contd.)

**Q3. What are grating lobes? How can they be avoided in array design?**

Grating lobes are secondary maximum beams that have the same amplitude as the main beam. They represent a waste of energy in undesired directions. They can be avoided by keeping the element spacing smaller than a wavelength ( $d < \lambda$ , or  $d < \lambda/2$  for steering).

**Q4. Why are binomial arrays designed? What is their main drawback?**

Binomial arrays are designed to completely eliminate sidelobes in their radiation pattern. Their main drawback is that they produce a wider main beam width (reduced resolution) and have lower directivity compared to a uniform array of the same length.

**Q5. Explain the key concept of the Dolph-Tschebyscheff array design.**

It uses Chebyshev polynomials to find the optimal balance between beamwidth and sidelobe level. It yields the narrowest possible main beam width for a specified, constant sidelobe level.

**Q6. Describe the basic idea of the Schelkunoff polynomial method.**

The array factor is written as a complex polynomial  $AF(z) = \sum a_n z^n$ . The nulls of the radiation pattern correspond to the roots of the polynomial that lie on the unit circle in the complex plane ( $|z| = 1$ ). Moving these roots allows shaping the pattern.

**Q7. How does a 2D planar array steer a beam in three dimensions?**

By using two independent progressive phase shifts  $\beta_x$  and  $\beta_y$  along the  $x$ - and  $y$ -axes, the phase can be matched to create a flat constructive wavefront in any direction  $(\theta_0, \phi_0)$ .

## Comparison of Linear Array Types

| Parameter             | Uniform ULA                         | Binomial Array                        | Dolph-Tschebyscheff                    |
|-----------------------|-------------------------------------|---------------------------------------|----------------------------------------|
| Excitation Amplitudes | Equal (1 : 1 : 1 : 1)               | Binomial coefficients (1 : 3 : 3 : 1) | Calculated using Chebyshev polynomials |
| Sidelobe Level        | High (first SLL $\approx -13.5$ dB) | No sidelobes (SLL = $-\infty$ )       | Constant SLL (set by designer)         |
| Directivity           | High                                | Low (broad beamwidth)                 | Optimum (narrowest beam for SLL)       |